

Tedlaye Tamiru Forsido

Tel: +27814705520

E-mail: tedlayef@yahoo.com

Personal Profile Statement

- A mature scientific research professional, with enduring capability to handle (from conceptualisation to delivery) execution of wastewater treatment researches and methods development with focus on cost, time and quality.
- Expert in environmental management, pollution protection and rehabilitation
- Sound experience on industrial and domestic wastewater as well as acid mine drainage treatment (neutralisation and heavy metal removal).
- Sound knowledge about regulations on water resources, care, reclamation and distribution.
- Expert in analytical and environmental chemistry including teaching and training
- Experimental techniques: IC/HPLC/AAS/ICP-OES

Education

2021 Doctorate of Technology Water Care: Tshwane University of Technology

Research Project

- Environmental science and pollution protection
- Waste Water Treatment Method Development
- Water Resource Management and Reclamation
- Resource Recovery
- Waste Management and Reuse
- Environment protection and sustainability education and training

2011 Masters of Technology in Analytical Chemistry: Tshwane University of Technology

Research Project

- Determination of fluoride content in environmental and biological samples
- Method development for determination of fluoride in environmental and biological samples

2000 Bachelor of Education in Chemistry

Relevant Modules

- General Chemistry
- Analytical Chemistry
- Pedagogy and Educational Psychology

Employment

2013-2018 Environmental Research Assistant

- Coordination and execution of projects within the research group
- Budgeting, scheduling, planning and execution of activities in projects
- Planning and initiation of cost effective substitute methods for industrial effluent and acid mine drainage treatment and pollutant removal projects
- Waste management, handling, valorisation, reuse and pollution mitigation
- Guide and participate in the development and implementation of environmental policy
- Procuring resources including personnel for projects
- Liaising with stakeholders
- Developing methods for the treatment of waste effluents and for recovery of valuable materials from solid industrial waste and sludge
- Performing analysis on various environmental samples, especially on solid and liquid industrial wastes
- Supervising and mentoring graduate and undergraduate students
- Educating and training water usage, management as well as environment protection and sustainability
- Undertaking environmental risk assessment and recommend ways of mitigation
- Reporting and publication of results

2008-2018 Lecturer

- Curriculum development
- Preparing annual educational manual, study materials and laboratory guides
- Preparing yearly budget for study materials, activities and utility services
- Lecturing analytical and environmental chemistry courses
- Providing practical laboratory education
- Organising guided study tours for students to various wastewater treatment plants

2002-2005 Quality Testing Expert

- Preparing and writing documents used in the quality assurance laboratory
- Guiding and managing technicians and personnel in the laboratory
- Checking and verifying the results from laboratory technicians
- Method validation
- Performing tests on the quality and standard of chemical, industrial and food products

2000-2002 Teacher

- Taught chemistry and other subjects at different levels of schools

1993-1996 Education Inspector

- Monitored the development activities of donor agencies.
- Did academic supervision to ensure the quality of education to students.
- Provided teachers training so that they can deliver quality teaching in classrooms.
- Communicated with donor and govt. agency informing about the progressive activity.
- Prepared weekly, monthly, periodical reports sending to concerned organisations.
- Supervised and managed a large number of office staff and their daily activity.
- Distributed education materials to government and private schools.
- Developed, coordinated and contributed to educational activity and delivered program.
- Carried out the administrative and communication tasks associated with educational activity and delivery program.
- Assisted with fundraising for the educational program, projects and events.
- Identified and developed online education opportunities.
- Prepared and distributed education newsletters.
- Developed and implemented a mentoring program.

Certifications

- Project management (**eLearning College**)
- Business Management (**Oxford Home Study Centre**)
- Climate Change: From Learning to Action (**UN CC Learn**)
- WASH Systems Strengthening: The Basics (**WASH Systems Academy**)
- Gender Equality and Human rights in Climate Action and Renewable Energy (**UN CC Learn**)
- The Building Blocks of WASH System (**WASH Systems Academy**)
- Finance for Sustainable Wash System (**WASH Systems Academy**)
- Environmental and Social Management System (**IDB**)
- Environmental Governance Certificate (**UNEP**)
- Introduction to the UN System: Orientation for Serving on a UN Field Mission (**Peace Operations Training Institute**)
- ISO 9000, ISO 17025

Skills

- Analytical & strategic planning
- Budgeting & forecasting
- Staff training & talent development
- Communication & interpersonal skills

Publications

- Neutralisation of acid effluent from steel manufacturing industry and removal of metals using an integrated electric arc furnace dust slag/lime process. *SN Applied Sciences*, (2019) 1:1605 | <https://doi.org/10.1007/s42452-019-1649-z>
- Removal of Al, Ba and Mg from industrial wastewater using EAFDS and lime, *Applied Water Science*, (2020) Article DOI: 10.1007/s13201-020-01234-y.
- Application of EAFDS/Lime Integrated System for the Removal of Cu and Mn from Industrial Effluent. *Int J Water Wastewater Treat* **6**(3): [dx.doi.org/10.16966/2381-5299.164](https://doi.org/10.16966/2381-5299.164)
- Characterization and Study of Different Electric Arc Furnace Steel Slags for Usage in Wastewater Treatment. *Jr. of Industrial Pollution Control* **37**(10)(2021) pp 1-4.
- Comparison of Different Total Ionic Strength Adjustment Buffer Compositions for Determination of Low Level Fluoride in Environmental Water Samples with Fluoride Ion Selective Electrode. *Environmental Systems Research*: <https://doi.org/10.21203/rs.3.rs-231958/v1>

References

- Mamo Muchie (Prof): DST-NRF SARChI Chair in Innovation Studies, Tshwane University of Technology, Email: muchiem@tut.ac.za/mammo.muchie@gmail.com
Tel: +27 12 382 3078
- Peter Ndibewu (Prof), Tshwane University of Technology, Senior Lecturer
Email: ndibewup@tut.ac.za, Tel: +2712 382 6300
- Dawit Kidanemariam (Dr): Postdoctoral Fellow, University of Pretoria.
Email: dawit.kidanemariam@up.ac.za, Tel: +27656464267